

Few-Shot Object Understanding and Grasp Proposals for Robotic Manipulation

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Abstract—Achieving robust robotic manipulation in unstructured environments requires precise 6-DoF grasp pose estimation for complex object geometries. To address this, we present an integrated pipeline leveraging RGB-D perception and semantic instance segmentation for multi-finger grippers. Target objects are isolated using SAM-3 in Promptable Concept Segmentation mode to generate precise instance masks directly from RGB input. These masks are fused with depth data and camera intrinsics to reconstruct object-level point clouds, which are denoised via geometric filtering to handle multiple-object scenes. To guide selection, a learned grasp-affordance module using a U-Net architecture predicts dense, per-pixel graspability heatmaps from aligned RGB streams. This heatmap serves as a 2D prior that constrains and guides grasp point selection before geometric candidate generation. Geometric descriptors from affordance-filtered point clouds generate candidate 6-DoF tripod grasps. Lightweight physical heuristics—evaluating center-of-mass, clearance, and collisions—then select robust poses for complex environments.

Index Terms—Few-shot perception, object understanding, multi-finger grasping, deep learning for visual perception

I. INTRODUCTION

Robotic grasping and manipulation remain fundamental challenges in autonomous robotics. A comprehensive review conducted by [1] highlights that, in practical applications such as warehouse automation and industrial handling, robotic systems must effectively detect, localize, and interact with objects under varying viewpoints and incomplete observations. Recent advances in robotic grasping have progressed from analytical formulations toward learning-based frameworks capable of performing in unstructured scenes. Early methods relied on geometric modeling, contact mechanics, and force-closure analysis, requiring precise object models and explicit grasp parameterization. A representative example is Dex-Net 2.0, where [2] demonstrated that robust grasp-quality

prediction can be learned directly from large-scale synthetic depth data and analytic grasp metrics. Few-shot and one-shot learning strategies have emerged as promising alternatives for reducing dependence on large annotated datasets. It has been demonstrated by [3] that robots can efficiently adapt to grasp new objects from only a limited number of examples, even in cluttered environments.

Vision models, combined with RGB-D sensing, support object-centric 3D scene representations by integrating semantic information extracted from RGB images with geometric information reconstructed from depth and point clouds [4], [5]. Methods such as GraspNet-1Billion [6] introduced large-scale benchmarks for grasp detection in cluttered scenes. In contrast, approaches such as VGN [7] showed that 3D scene representations can be used to predict grasp pose, orientation, and gripper configuration directly from volumetric input. These advances in 6-DoF grasp synthesis have also increased the importance of robust visual perception in robotic manipulation pipelines. Accurate grasp generation in cluttered scenes depends not only on geometric reasoning from point clouds but also on reliable object localization and scene understanding from visual input.

More recent work has explored graspability estimation as an intermediate representation before grasp generation. Graspness Discovery [8] introduced point-level graspability representations for identifying graspable regions in point clouds. Similarly, approaches such as HGGD [9] and FlexLoG [10] use graspability maps to guide candidate generation toward regions with high grasp potential before geometric evaluation.

Building on these developments, promptable segmentation models have become increasingly relevant for object-aware robotic grasping. Segment Anything [4] enables accurate target localization directly from RGB images, allowing grasp reasoning to focus on the selected object. Extending this idea, [11] proposed GraspSAM, integrating segmentation and grasp prediction into a unified framework.

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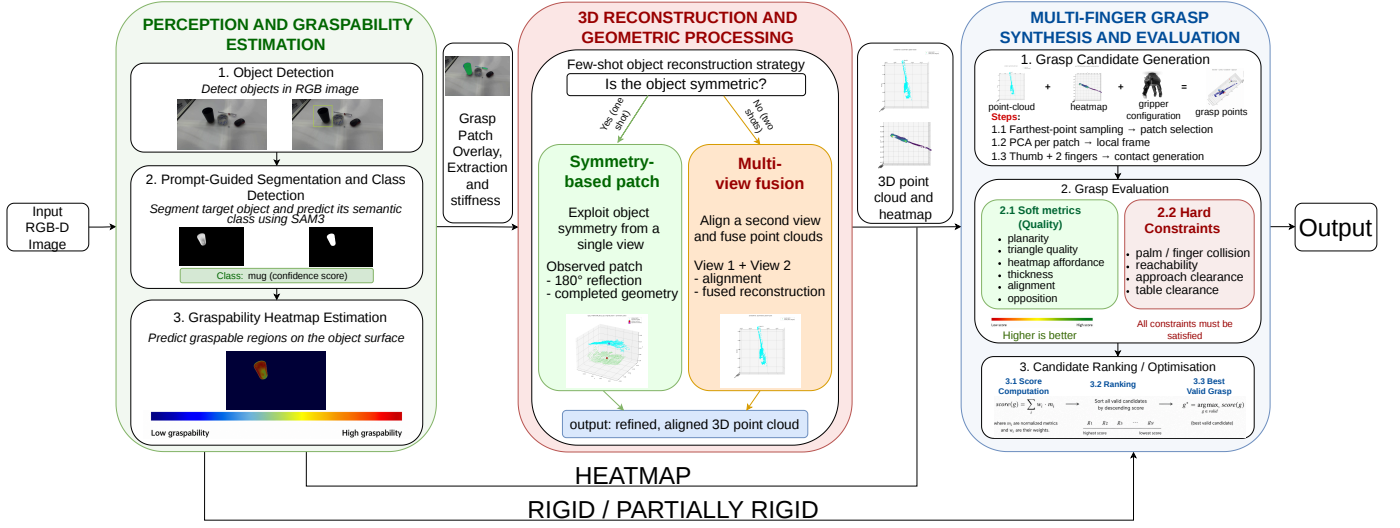


Fig. 1. Few-Shot Architecture: Overview of the proposed semantic-guided robotic grasping pipeline integrating RGB-D perception, object-centric segmentation, graspability estimation, 3D reconstruction, and multi-finger grasp synthesis.

This work proposes an architecture that combines object detection, segmentation, 3D reconstruction, and grasp generation in a unified pipeline for few-shot object understanding and grasp pose estimation from RGB-D data. Semantic information is used to identify the target object and estimate graspable regions, while geometric processing is used to reconstruct the object surface and generate candidate grasp configurations. These candidates are then evaluated and ranked using local geometric features, contact-triangle structure, and feasibility constraints specific to a three-finger gripper. The main objective of this research is to investigate how semantic perception and geometric reasoning can be integrated to improve object-centric grasp generation in RGB-D environments. The proposed framework aims to improve grasp robustness under partial observations, occlusions, and limited training examples.

The main contributions of this paper are :

- the development of a unified semantic-guided grasping pipeline integrating object detection, prompt-guided segmentation, graspability estimation, 3D reconstruction, and multi-finger grasp synthesis from RGB-D observations;
- a graspability-driven object reconstruction framework that pairs symmetry-based patch completion with multi-view fusion for geometric refinement;
- the implementation of a geometric grasp candidate generation and evaluation framework based on local surface descriptors and gripper-specific feasibility constraints for a three-finger robotic gripper.

The paper is organized as follows: Section II formulates the problem and framework; Sections III–V detail perception, 3D reconstruction, and grasp candidate selection; Section VI presents experimental results; and Section VII concludes with future directions.

II. PROBLEM FORMULATION AND SYSTEM OVERVIEW

The proposed framework follows a modular, semantically guided robotic grasping pipeline that enables robust grasp gen-

eration for previously unseen objects in multiple-object RGB-D scenes. It integrates perception, graspability estimation, 3D reconstruction, and multi-finger grasp synthesis from RGB-D observations, as illustrated in Fig. 1.

Given an RGB-D image, the perception stage first operates on the RGB component to obtain an object-level region of interest. A catalog-constrained zero-shot detector is used to search only for objects in the GraspNet-derived object dictionary, which maps each internal object key to a canonical object name, a set of detector labels, a SAM 3 prompt, a symmetry label, and a flexibility label used in the later reconstruction stage. In practice, the detector provides candidate object names and bounding boxes, while the associated catalog entry provides the text prompt passed to SAM 3, which generates a binary mask for the selected object, which is used to suppress the background before graspability estimation. This design keeps object localization, segmentation, and graspability prediction modular: the zero-shot detector selects *what* object should be processed, SAM 3 determines *where* the object is in the image, and the heatmap network estimates *where on the object* grasping is likely to be successful [12], [13].

The segmented RGB image is then processed by the graspability estimation module, which predicts a dense heatmap over the object’s visible surface. A U-Net-based [14] encoder-decoder architecture is used to estimate soft graspability values, where each pixel receives a score indicating its suitability for grasping. The network is trained using a compound sparse-heatmap loss defined as

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{focal}} + \lambda_1 \|\hat{H} - H\|_1 + \lambda_d \mathcal{L}_{\text{dice}}, \quad (1)$$

where H is the generated graspability matrix shaped target and $\hat{H}$ is the predicted heatmap. The $\mathcal{L}_{\text{focal}}$ loss term reduces the influence of easy background pixels and focuses training on difficult or sparse positive regions [15]; the $\|\cdot\|_1$ loss term preserves the continuous intensity values of the heatmap;

and the $\mathcal{L}_{\text{dice}}$ loss term encourages spatial overlap between predicted and target graspable regions, which is useful when the positive regions occupy a small part of the image [16].

The segmentation mask and graspability heatmap are then forwarded to the reconstruction stage together with the aligned depth map. The mask is applied to the depth image to isolate the target object, while the heatmap is used to localize high-graspability surface regions for further geometric processing. These regions are projected into 3D using the camera intrinsic parameters. For a pixel $p = (u, v)$ with depth $d = D(u, v)$, where D is the depth map, the corresponding 3D point is computed through back-projection as

$$z = d, \quad x = \frac{(u - c_x)z}{f_x}, \quad y = \frac{(v - c_y)z}{f_y}, \quad (2)$$

resulting in an object-level point cloud representation of the target surface.

To address incomplete geometry arising from occlusions or single-view acquisition, the reconstruction strategy is selected based on the object's symmetry. For approximately symmetric objects, local patch completion is achieved via symmetry-based reconstruction from a single view, reflecting the observed surface about the dominant symmetry axis. This strategy was applied, for example, to cylindrical objects such as cups, where the missing geometry can be captured from a single image. For asymmetric objects, geometric completion is performed through multi-view fusion. In this case, an additional RGB-D image is acquired and aligned with the initial reconstruction via point cloud registration, thereby merging complementary object surfaces into a unified representation. This strategy was used for asymmetric objects such as adhesive tape rolls or irregular mechanical parts, where a single view does not provide sufficient geometric coverage. The resulting refined object-level point cloud is then used in the grasp reasoning stage.

In this stage, we use the reconstructed object point cloud X and the generated affordance heatmap. The grasp synthesis problem is defined as selecting the valid candidate γ with the highest final score

$$g^* = \arg \max_{g \in \gamma} S(g), \quad (3)$$

where $S(g)$ denotes the final grasp quality score, g^* is the grasp that has the highest score, and $\gamma \subseteq \Gamma$, where Γ is the total collection of valid candidates that satisfy geometric and gripper feasibility constraints, including contact stability, reachability, and compatibility with the opening limits of the underactuated three-finger gripper.

In the final stage, the reconstructed point cloud, together with the graspability scores propagated from the predicted heatmap, is used to synthesize and evaluate multi-finger grasps. Local grasp patches are sampled from the reconstructed surface, and candidate contact configurations are generated around regions with high graspability response. Each grasp candidate is then evaluated using geometric quality metrics, including local surface planarity, contact-triangle quality, alignment with the dominant object direction, local object

thickness relative to the gripper opening, and heatmap-based grasp affordance. In parallel, feasibility constraints such as contact stability, collision avoidance, rigidity, and reachability are applied to reject physically invalid grasps. The remaining valid candidates are ranked by final grasp score, and the highest-scoring grasp configuration is selected as the pipeline output.

III. PERCEPTION AND IMAGE GRASPABILITY ESTIMATION

The graspability dataset was generated from GraspNet-style grasp annotations in order to train a dense model that predicts image regions likely to support a successful grasp. The procedure follows the intuition of graspness-based methods, where grasp detection is decomposed into a first stage that estimates *where* grasping is promising and a second stage that estimates *how* the grasp should be executed [8]. In this work, the generated heatmaps are used only as intermediate graspability supervision. They do not directly represent final grasp poses, but rather guide the later grasp selection stage toward promising object regions.

For each RGB-D frame, the annotation file contains a set of planar grasp rectangles. Each grasp annotation is represented as

$$g_i = (x_{1,i}, y_{1,i}, x_{2,i}, y_{2,i}, h_i, q_i, o_i), \quad (4)$$

where $(x_{1,i}, y_{1,i})$ and $(x_{2,i}, y_{2,i})$ define the grasp axis in the image plane, h_i defines the thickness of the grasp rectangle around this axis, q_i is the grasp quality score associated with the annotation, and o_i is the identifier of the object to which the grasp belongs. The two endpoints determine the position, orientation, and projected opening of the grasp, while h_i provides the rectangle's second dimension. The score q_i determines the intensity contribution of the grasp in the generated heatmap, which is later normalized to the interval $[0, 1]$. The object identifier o_i is used only during dataset generation to constrain each grasp rectangle to its corresponding object mask; this avoids assigning graspability to background pixels or to neighbors in multi-object scenes

$$c_i = \left(\frac{x_{1,i} + x_{2,i}}{2}, \frac{y_{1,i} + y_{2,i}}{2} \right), \quad (5)$$

$$w_i = \sqrt{(x_{2,i} - x_{1,i})^2 + (y_{2,i} - y_{1,i})^2},$$

$$\theta_i = \text{atan2}(y_{2,i} - y_{1,i}, x_{2,i} - x_{1,i}). \quad (6)$$

Thus, each annotation becomes a rotated image-space grasp rectangle $r_i = (c_i, w_i, h_i, \theta_i)$ with an associated score q_i .

The heatmap is generated by rasterizing all valid grasp rectangles into the image plane. For each pixel p , two maps are accumulated: a score map $S_Q(p)$ and a support map $C(p)$:

$$S_Q(p) = \sum_i q_i \mathbf{1}[p \in r_i], \quad C(p) = \sum_i \mathbf{1}[p \in r_i]. \quad (7)$$

Here, $\mathbf{1}[p \in r_i]$ is equal to one when pixel p lies inside the rotated grasp rectangle r_i , and zero otherwise. The score map stores the accumulated grasp quality at each pixel, while the support map counts how many grasp annotations cover that pixel.

A first graspability estimate is obtained by averaging the accumulated scores:

$$\bar{Q}(p) = \frac{S_Q(p)}{\max(C(p), \epsilon)}. \quad (8)$$

This avoids assigning high values to a pixel only because it is covered by many overlapping rectangles. However, pixels supported by several consistent grasp annotations should still be considered more reliable than pixels covered by only one annotation. For this reason, a bounded support weighting term is applied:

$$A(p) = 1 - \exp\left(-\frac{C(p)}{\tau}\right), \quad (9)$$

where τ controls how quickly the support contribution saturates. The raw graspability heatmap is then defined as

$$H_{\text{raw}}(p) = \bar{Q}(p)A(p), \quad (10)$$

where $H_{\text{raw}}(p)$ is the raw graspability score associated to pixel p_i . This produces a stable, dense target in which high values correspond to pixels that are both associated with high-quality grasp annotations and supported by sufficient annotation evidence.

When object labels are available, the rasterized grasps are constrained by the object mask associated with their object identifier. This prevents grasp rectangles from contributing to background pixels or to neighbors in multi-object scenes. A global foreground mask can also be applied so that the final heatmap contains positive values only inside labeled object regions. This masking step is especially important because the generated heatmap is later used to train an object-centric graspability estimator.

An optional refinement step is applied using geometric cues from the depth map. First, high-confidence heatmap locations are selected as seeds, and their responses are then propagated to nearby pixels with similar depth and surface-normal direction. In addition, a Canny edge map is computed from the normalized depth image and used to highlight regions near depth discontinuities. This stage can be interpreted as a depth-edge-guided heatmap refinement step: it preserves the annotation-derived graspability map while encouraging high-response regions to align more closely with local 3D structure and object boundaries.

Finally, the heatmap is smoothed and robustly normalized before being saved as an 8-bit PNG image. Let P_l and P_h be the lower and upper percentile values used for robust normalization. The final target map is computed as

$$H_{\text{out}}(p) = \text{clip}\left(\frac{H(p) - P_l}{P_h - P_l + \eta}, 0, 1\right), \quad (11)$$

where H denotes the final heatmap after optional refinement, P_l and P_h are the lower and upper percentile values used for robust normalization, and η is a small constant that prevents division by zero. The operator $\text{clip}(\cdot, 0, 1)$ truncates all values below 0 to 0 and all values above 1 to 1.

This ensures that the final heatmap is a valid, normalized target with values in $[0, 1]$ before being converted to an 8-bit PNG image.

IV. OBJECT-LEVEL 3D RECONSTRUCTION AND GEOMETRIC PROCESSING

This section describes the reconstruction strategy used to generate an object-level 3D representation from RGB-D observations. Using the segmentation mask and graspability heatmap obtained in the previous stage, the module reconstructs and refines the local object geometry required for grasp candidate generation.

Starting from the object regions, the reconstruction module generates a local 3D representation of the target surface from the aligned depth observation. The selected pixels are projected into 3D using the intrinsic camera parameters as defined in Equation (2), yielding an object-level point cloud representation of the visible surface. Local interpolation is then applied around the selected graspable patches to obtain a denser, more continuous surface representation suitable for further analysis.

To compensate for missing geometry due to partial observations and occlusions, the reconstruction strategy is adapted to the object's symmetry.

For symmetric objects, local completion is performed directly on the selected graspable patches by reflecting the observed surface points around the dominant symmetry axis

$$\text{patch}_i^{\text{sym}} = S_\Phi \text{patch}_i, \quad (12)$$

where S_Φ denotes the symmetry transformation matrix.

Here, patch_i denotes the local 3D patch represented as a set of 3D points extracted from the object point cloud, and the symmetry transformation is applied independently to each point in the patch.

For asymmetric objects, geometric completion is achieved through multi-view fusion. A second RGB-D observation is acquired and aligned with the initial view using a rigid point cloud registration procedure inspired by ICP-based alignment methods [17]. The alignment between the two observations is defined as:

$$\text{patch}'_i = R \text{patch}_i + t \quad (13)$$

where R and t represent the estimated rotation and translation between the two observations.

After alignment, the reconstructed geometry is obtained through point cloud fusion:

$$pc_{\text{fused}} = pc_1 \cup T(pc_2) \quad (14)$$

where $T(pc_2) = R pc_2 + t$ denotes the rigid transformation applied to the second-view point cloud before fusion.

The reconstructed point cloud is further refined through voxel downsampling and statistical outlier removal to suppress noise and remove isolated measurements. For each point b , the average distance to its local neighborhood is computed as:

$$\text{dist}_i = \frac{1}{k} \sum_{j=1}^k \|b_i - b_j\| \quad (15)$$

and points exceeding the defined threshold are removed from the reconstruction.

After refinement, geometric descriptors are extracted in order to characterize the spatial structure and orientation of the reconstructed object. Principal component analysis (PCA) is applied through the covariance matrix

$$\Sigma = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X})^T \quad (16)$$

where X_i denotes the reconstructed 3D points and $\bar{X}$ represents the centroid of the reconstructed point cloud. The resulting principal directions provide local geometric information used in the next stage for grasp patch sampling, contact generation, and grasp evaluation.

The output of the 3D reconstruction phase is a refined object-level point cloud enriched with geometric descriptors that capture the dominant spatial orientation of the target object. This representation is forwarded to the multi-finger grasp synthesis module, where candidate grasps are generated and evaluated according to geometric quality and feasibility constraints.

V. GRASP CANDIDATE GENERATION AND SELECTION

Following object reconstruction, grasp candidates are generated directly on the surface. The grasping strategy extends an analytical geometric baseline with a heatmap-derived semantic prior and a flexibility term for deformable objects. In the baseline setup, the entire point cloud acts as a global search region. PCA establishes a local reference frame using the patch centroid as the origin, two dominant eigenvectors as tangent directions, and the smallest eigenvector as the surface normal. Based on this frame, grasp candidates are generated using a one-thumb and two-fingers opposed configuration adapted to the Robotiq 3-Finger gripper [18], producing a triangular contact structure suitable for underactuated three-finger grasping [19]. The three contacts consist of one thumb (F_1) opposed by two fingers (F_2, F_3) offset symmetrically at $\pm \frac{\phi}{2}$ along the object's principal axis. The pipeline evaluates all four Robotiq 3F gripper modes (basic, wide, pinch, scissor) and multiple finger-spread values for each mode, so the gripper aperture adapts to the local object geometry without requiring explicit shape knowledge.

Valid contact candidates are restricted to surface points that face the camera and lie within a thin slab around the local tangent plane. Contact candidates in the lowest portion of the object height are additionally excluded to prevent grasping from below. Fingers are placed at the furthest perpendicular edge within a longitudinal window, favoring the object's visible margin over its smooth interior

Each candidate is evaluated using several geometric quality measures, including contact triangle quality, opposition between contact normals, local object thickness relative to the optimal gripper aperture range, opening compatibility, approach alignment with the object geometry, and local surface planarity. The alignment term is modulated by the estimated object elongation, such that approach direction becomes in-

creasingly important for elongated objects while remaining largely unconstrained for near-isotropic shapes

$$s_{\text{align}} = \varepsilon \cdot (1 - |a \cdot ax|) + (1 - \varepsilon), \quad (17)$$

where a denotes the approach direction, ax the principal object axis, and $\varepsilon \in [0, 1]$ the elongation factor. Consequently, approaches perpendicular to the principal axis are preferred for elongated objects, whereas all approach directions receive similar scores when the object is approximately isotropic. Hard constraints reject candidates that fail minimum triangle quality, minimum opposition, opening limits, diameter compatibility, or contact reachability requirements.

When graspability heatmaps are available, the heatmap influences the contact selection within each edge band and contributes an additional semantic term to the final score.

For flexible objects, the ranking function includes an additional flexibility term that can be 0 if the object is flexible or 1 if it is rigid. Instead of using the measured local diameter ϕ , the evaluation is performed with an effective diameter

$$\phi_{\text{eff}} = d(1 - r_c), \quad (18)$$

where $r_c \in [0, 1]$ is a fixed compression ratio that approximates the expected deformation during grasp closure. The effective diameter d_{eff} is used when computing the thickness compatibility score and the diameter feasibility constraints. As a result, grasp candidates that slightly exceed the nominal diameter limits can still receive favorable scores if the object is expected to compress during grasping. This flexibility term is applied only when the object is classified as flexible and is added on top of the existing scoring function, leaving the baseline and heatmap-guided variants unchanged.

The final grasp score is a normalized weighted combination of the active terms,

$$S(g) = \frac{\sum_k w_k s_k(g)}{\sum_k w_k}, \quad (19)$$

where $w_k \in [0.08, 0.24]$ are fixed weights, $s_k(g)$ is an individual quality term from:

- s_{Δ} (*triangle quality*) evaluates the symmetry and stability of the triangle formed by the three contact points.
- s_{opp} (*opposition*) measures how well the contact normals oppose each other to generate stable grasping forces.
- s_{thick} (*thickness*) evaluates whether the local object thickness lies within the feasible gripper closing range.
- s_{open} (*opening comfort*) favors grasps requiring a smaller finger opening relative to the gripper aperture.
- s_{align} (*alignment*) measures how well the grasp orientation matches the principal object geometry.
- s_{plan} (*planarity*) evaluates local surface flatness around the contact region.
- s_{hm} (*heatmap*), when enabled, incorporates visual affordance information predicted by the network.

The highest-scoring valid candidate is selected as the output grasp.



Fig. 2. Representative examples used during graspability heatmap training. From left to right: RGB training image, corresponding ground-truth graspability heatmap, testing RGB scene, and object-specific heatmap responses for the bottle and cream containers.

VI. RESULTS AND DISCUSSIONS

This section delivers a comprehensive evaluation of the proposed few-shot grasping architecture, validating overall system performance and delineating the algorithmic contributions of its constituent modules. The validation pairs quantitative benchmarks with qualitative assessments across a diverse suite of representative target objects. The analysis highlights the performance impacts of the learned graspability heatmap and the flexibility-aware scoring mechanism, and characterizes their exact influence on the final grasp selection policy.

These initial experiments validate the proposed concept and demonstrate the graspability heatmap’s contribution to the overall pipeline. However, thoroughly evaluating generalization requires iterative testing across a larger set of objects with repeated acquisitions and statistical analysis of the heatmaps. Such an evaluation will provide stronger evidence regarding the consistency, robustness, and repeatability of the learned visual affordance representation.

The dataset consists of 100 training scenes, each containing 256 images, resulting in a total of 25 600 training images and 72 distinct object instances. The validation and test sets were generated from 10 previously unseen scenes each, with 2 560 images per split. Table I summarizes the default architecture and training configuration of the graspability heatmap network. The model follows a U-Net-style encoder-decoder design with a ResNet-34 backbone and produces a single-channel graspability heatmap.

The network is trained with a compound sparse-heatmap loss composed of focal binary cross-entropy, mean absolute error regression, and soft dice loss-equation (1). In the implementation, the weight values are:

$$\lambda_f = 1.0, \quad \lambda_l = 0.5, \quad \lambda_d = 0.25.$$

TABLE I
PARAMETERS OF THE GRASPABILITY HEATMAP NETWORK

Component	Value
Backbone	ResNet-34
Pretraining	ImageNet-1K weights
Encoder channels	[64, 64, 128, 256, 512]
Decoder channels	(256, 128, 64, 32)
Upsampling	Bilinear interpolation
Activation	SiLU
Normalization	Batch Normalization
Dropout	0.1
Output channels	1
Output type	Graspability logits
Final activation for heatmap	Sigmoid
No. epochs	10

The focal term is computed on logits, while the L1 and Dice terms are computed after applying the sigmoid activation to obtain the predicted heatmap.

Figure 2 illustrates representative examples from the graspability heatmap training pipeline. The first two images show an RGB training sample together with its corresponding ground-truth heatmap annotation used for supervision. The remaining examples present a reference RGB scene and the resulting object-specific heatmap responses generated for selected objects. These visualizations highlight how the network learns to associate image regions with graspable areas and provide pixel-level graspability predictions that are later integrated into the grasp candidate ranking stage.

The grasping stage was evaluated in three configurations: *Baseline*, where grasp candidates are generated and ranked only from geometric cues; *+Heatmap*, where the same pipeline is augmented with the predicted graspability prior; and *+Heatmap+Flexibility*, where a flexibility scoring term is additionally enabled for deformable objects. The final score in all cases corresponds to the normalized weighted ranking defined in Section V.

1) Heatmap Guided vs. Geometry Only Point Cloud:

This evaluation compares the geometry only baseline vs. the heatmap-guided approach in order to assess whether the predicted graspability map provides a useful visual prior for grasp selection. Two representative objects were analyzed: a cup (symmetric) and a screwdriver (asymmetric). For each object, the grasping stage was executed in two configurations: *Baseline*, and *+Heatmap*. The numerical results are summarized in Table II.

Table II summarizes the quantitative comparison between the geometry-only baseline and the heatmap-guided configuration for two representative objects. For each case, the final grasp score, heatmap contribution, selected grasp mode, and number of evaluated candidates are reported. The results show that the heatmap prior improves both the final score and candidate diversity for the cup while having a limited

TABLE II
EVALUATION 1: BASELINE VS. HEATMAP GUIDED GRASPING

Object	Conditions	Score	Heat	Mode	Candidates
Blue cup	Baseline	0.5235	0	WIDE	12
Blue cup	+Heatmap	0.5865	0.9048	WIDE	58
Screwdriver	Baseline	0.6891	0	PINCH	12
Screwdriver	+Heatmap	0.5846	0.0044	PINCH	12

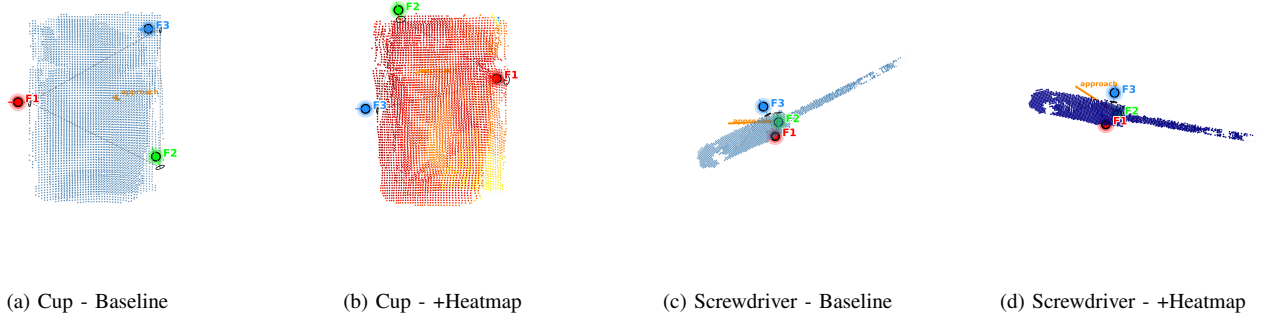


Fig. 3. Qualitative comparison between the geometry-only baseline and the heatmap-guided variant. (a,b) Cup grasps before and after adding the heatmap prior, which improves candidate selection and final scoring. (c,d) Screwdriver grasps for the same comparison.

influence in the screwdriver example due to the weak heatmap response. For the cup, the heatmap prior improves the final score from 0.5235 to 0.5865 while increasing the number of evaluated candidates from 12 to 58. This result suggests that the visual prior expands the search space toward additional highly graspable contact regions, rather than simply changing the ranking of the geometric candidates. As illustrated in Fig. 3, the resulting grasp is also relocated to a more favorable contact region on the object surface. The fixed weights are $w_{\Delta} = 0.24$, $w_{\text{thick}} = 0.2$, $w_{\text{opp}} = 0.2$, $w_{\text{hm}} = 0.2$, $w_{\text{open}} = 0.14$, $w_{\text{align}} = 0.12$, $w_{\text{plan}} = 0.08$, with w_{hm} included only when the heatmap prior is active.

Overall, the results show that the graspability heatmap provides an effective visual prior for grasp selection. When reliable visual cues are present, they improve candidate exploration and lead to better final grasp configurations. However, when the heatmap response is weak, as in the screwdriver example from Table II, the additional term can reduce the normalized score because its weight remains included despite the negligible heatmap contribution.

2) *Effect of the Flexibility Aware Term*: This evaluation focuses on the contribution of the flexibility-aware scoring term, which is activated after the heatmap-guided configuration has already been introduced. The purpose of this experiment is to determine whether a simple local compression model can improve grasp ranking on objects that may deform during gripper closure.

In the current implementation, flexibility is modeled by reevaluating the local thickness term under a 15% (r_c in Equation 18 is set to 0.15) diameter reduction, which approximates a small, uniform compression during grasp closure. This extension is only used during the scoring stage. Therefore, it does not modify the reconstructed point cloud, the generated candidate set, or the contact placement strategy, but only the final ranking assigned to the candidate grasps.

A representative example is provided by the lotion bottle. Two configurations were compared: *+Heatmap*, where the graspability prior is enabled but no flexibility term is used in the final score, and *+Heatmap+Flex*, where the flexibility aware term is added on top of the heatmap guided score.

TABLE III
EVALUATION 2: EFFECT OF THE FLEXIBILITY AWARE SCORING TERM

Object	Conditions	Score	Heat	Flexibility	Mode
Hand cream tube	+Heatmap	0.7899	0.7964	0	WIDE
Hand cream tube	+Heatmap+Flex	0.8063	0.7964	1	WIDE
Lotion bottle	+Heatmap	0.6340	0.6887	0	BASIC
Lotion bottle	+Heatmap+Flex	0.7264	0.6887	1	BASIC



(a) Lotion bottle: +Heatmap (b) Lotion bottle: +Heatmap+Flex

Fig. 4. Qualitative comparison for the lotion bottle under *+Heatmap* and *+Heatmap+Flex* configurations. Enabling the flexibility-aware scoring term shifts the selected grasp toward a contact region more compatible with the compressed diameter.

The numerical results are summarized in Table III, and the qualitative comparison is shown in Fig. 4. Table III summarizes the quantitative comparison between the heatmap-guided configuration and the heatmap-guided configuration with the flexibility-aware scoring term enabled. For each object, the final grasp score, heatmap contribution, flexibility term, and selected grasp mode are reported. In both examples, enabling the flexibility term yields higher final scores, indicating that local deformation awareness can improve grasp evaluation and final grasp selection for compliant objects.

For the lotion bottle, enabling the flexibility-aware term leads to a different grasp selection. The contact locations shift toward a region whose effective compressed diameter falls within the admissible gripper closing range. This results in an increase of the final score from 0.6340 to 0.7264. This shows

that the flexibility term can influence both candidate ranking and the final grasp selection by making previously penalized grasps feasible.

The results shown in Table III and Fig. 4 are a representative visualization of the flexibility term. By replacing the nominal diameter with an effective compressed diameter, the scoring function prioritizes candidates that would otherwise be penalized under the rigid-object assumption. When such candidates achieve a higher overall score than the heatmap-guided solution, a different grasp is ultimately selected. Together with the hand cream tube experiment, these results indicate that flexibility-aware scoring can meaningfully affect grasp selection for deformable objects, provided that the assumed compression model adequately reflects the object's actual mechanical behavior.

A testing robotic system including a UR5 manipulator, a three-finger gripper, and an eye-in-hand stereo camera was used to evaluate the framework design. Starting from an RGB-D scene, the pipeline performs object segmentation, graspability heatmap prediction, point cloud reconstruction, contact point generation, and grasp candidate ranking. The highest-scoring grasp configuration is then sent to the robotic manipulator for execution. This experiment validates the integration of all pipeline components under real operating conditions, from visual perception to physical grasp execution. A video recording of the complete experiment is available online at: https://youtu.be/aOvr3jT-G_g. The end-to-end experiment confirms that the proposed pipeline can be successfully deployed in a real grasping scenario. The system was able to generate feasible grasp candidates directly from RGB-D input and execute the selected grasp.

VII. CONCLUSIONS

This paper presented a semantic-guided robotic grasping pipeline for few-shot object understanding and grasp proposal generation from RGB-D observations. The proposed framework combines prompt-guided object segmentation, graspability heatmap estimation, object-level 3D reconstruction, and multi-finger grasp synthesis within a unified perception-to-action architecture. By explicitly embedding semantic visual priors into geometric grasp reasoning, the method enables robust, data-efficient grasp generation for previously unseen objects in unstructured environments.

Experimental results demonstrated that the proposed approach generates stable and feasible grasp configurations across a diverse suite of object categories. The heatmap-guided evaluation verified that the predicted graspability prior significantly accelerates candidate exploration and guides selection toward optimal contact points when reliable visual cues are available. Additionally, the flexibility-aware scoring term improved grasp evaluation for highly deformable objects by explicitly accounting for local physical compression during grasp closure. Real-world robotic experiments further validated the framework design, confirming seamless execution from raw RGB-D perception to physical object manipulation.

Future work will address physical experiment limitations (such as lighting sensitivity, calibration errors, and latency),

while expanding the framework to deformable, symmetric, and complex geometries or heterogeneous materials. Additionally, we plan to improve graspability prediction via data augmentation and conduct extensive, long-horizon manipulation trials to evaluate overall operational robustness.

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